Model parameters	1	2	3	4	5	6	7	8
1 - HKY Strict Constant	-	<1	<1	<1	<1	<1	<1	<1
2 - HKY Relaxed Exp.	>150	-	<1	<1	<1	<1	<1	<1
3 - HKY Strict Exp.	>150	20-150	-	<1	<1	<1	<1	<1
4 - GTR Strict Expansion	>150	>150	>150	-	<1	<1	<1	<1
5 - GTR Relaxed Exp.	>150	>150	>150	>150	-	<1	<1	<1
6 - GTR Strict Exp.	>150	>150	>150	>150	>150	-	<1	<1
7 - GTR Relaxed Skygrid*	>150	>150	>150	<150	>150	20-150	-	<1
8 - GTR Strict Skygrid*	>150	>150	>150	>150	>150	>150	>150	-

HKY/GTR *- Nucleotide substitution model*
Strict/Relaxed *- Clock model*

Constant/Exponential/Expansion/Skygrid *- Coalescent tree prior*
**Our analyses indicated that a structured Bayesian Skygrid coalescent with a strict clock was the best-supported model. However, due to the significant computational demands of running a Skygrid model across all lineage and phylogeographic model combinations, we opted for an exponential coalescent model with a strict clock. This approach remains appropriate, as our analysis focused on the early spread and origins of each lineage, with sequences selected from the initial phase of their respective epidemic waves. This selection reduces the effect of later population structure while still capturing key transmission dynamics.*